WEAK PRE-ADICNESS DESCENDS ALONG A TAUT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement opens with the chapter's phrase 'In the situation above', which fixes the two rings and the taut map between them; that situation is recorded as the first standard prerequisite and is referenced by label, no context span being delivered for it.

Editorial note. The proof displays one containment of ideals; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The characterisation of weakly pre-adic rings used here. Same chapter, Lemma lemma-weakly-adic, invoked at proof lines 1862-1864. Reduces weak pre-adicness to pre-admissibility together with openness of the closures of the powers of an ideal of definition.
- Pre-admissibility descends to the source along the taut map. Same chapter, Lemma lemma-taut-descent-admissible, invoked at proof lines 1864-1866. Gives the first half of the characterisation for the source.
- The closure of the extension of an ideal is the intersection of its images in the quotients. Same chapter, Lemma lemma-closed, invoked at proof lines 1876-1878. Identifies the closure appearing in the argument as an intersection, which is what makes the containment testable modulo each open ideal.
- An inclusion of ideals can be tested after a faithfully flat extension. Standard, used at proof lines 1884-1888. Descends the containment of the images in the quotients from B to A.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GXN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1839–1839. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1633–1633. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *In the situation above, if B is weakly pre-adic, then A is weakly pre-adic.*

Proof. We will use the characterization of weakly pre-adic rings given in Lemma 0GXF without further mention. By Lemma 0GXN the topological ring A is pre-admissible. Let $I \subset A$ be an ideal of definition. Fix $n \geq 1$. To prove the lemma we have to show that the closure of I^n is open. Let $I_\lambda \subset A$ be a fundamental system of open ideals. Denote $J \subset B$, resp. $J_\lambda \subset B$ the closure of IB, resp. $I_\lambda B$. Since B is weakly pre-adic, the closure of J^n is open. Hence there exists a λ such that

$$J_\lambda \subset \bigcap\nolimits_\mu (J^n + J_\mu)$$

because the right hand side is the closure of J^n by Lemma 0AMS. This means that the image of J_λ in B/J_μ is contained in the image of J^n in B/J_μ. Observe that the image of J^n in B/J_μ is equal to the image of I^nB in B/J_μ (since every element of J is congruent to an element of IB modulo J_μ). Since $A/I_\mu \to B/J_\mu$ is faithfully flat and since $I_\lambda B \subset J_\lambda$, we conclude that the image of I_λ in A/I_μ is contained in the image of I^n. We conclude that I_λ is contained in the closure of I^n and the proof is complete. $\square$